uOttawa

UNIVERSITÉ D'OTTAWA · UNIVERSITY OF OTTAWA

Week 3 — Training Day 2: Mission Operations Training

ORBIT NORTH · Tue 18 August 2026 · uOttawa Cyberrange

ORBIT NORTH
Summer 2026

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

GNG 3100 · SYS 5186 · ORBIT NORTH · Issue 1.0 · 2026-10-30

Classification: Internal

Training Day 2 — Mission Operations Training

Day at a glance

When	What	Output
09:00 – 10:00	Voice-loop discipline + GO/NO-GO protocol	Voice card learnt
10:00 – 11:30	Procedure walk-through: NOM-001 and CTG-003	Q&A captured
11:30 – 12:00	Shift handover practice (NOM-012)	Two handovers logged
13:00 – 14:00	Flight-dynamics workshop	Pass timing predicted
14:00 – 15:30	Dry-run 1 — nominal pass, no failures	Operator-shift logs
15:30 – 16:30	Dry-run 2 — nominal pass with one failure injected	Anomaly tickets, debrief
16:30 – 17:00	**Operations training quiz**	Quiz submitted

Setting: uOttawa Cyberrange. Two consoles set up by 09:00. Both dry-runs use the **Nominal Post-Commissioning** breakpoint (BP-1).

---

1. Voice-loop discipline

Textbook Part 2, §§9.1–9.2, is the session text and the reference card: the addressing pattern ("FD, AOCS — go for star-tracker enable": the station called, then the caller, then the message), the GO/NO-GO call-and-response, the hot-mike rule (push-to-talk; only the FD goes hot-mike, and announces it), loop discipline, and the rule that the FD authorises while the owning position commands. Run the three drills of §9.2: addressing, GO/NO-GO with a hold, and an anomaly callout.

The callouts stay in English for every crew. The French edition of the textbook gives the meaning of each callout in French in the same table (§9.2, *Callout vocabulary*), which serves as the EN/FR glossary.

1.1 Standard callouts

The callout table is textbook Part 2, §9.2, *Callout vocabulary*: 15 callouts with meaning and expected response, from "Comm check 1-2-3" to "Loop discipline", including "Standby plus 30", "AOS in N" and "Pass closed". Print it as the voice card.

1.2 GO/NO-GO protocol

Before any command that changes the spacecraft's state, the FD polls the positions concerned ("AOCS, EPS, FDIR — go for ST1_POWER?"). Each answers in poll order: *Go*, *No-go, [reason]* or *Hold, [reason]*. The FD decides and declares it on the loop: "Go for ST1_POWER" or "Hold on ST1_POWER, [reason]". The owning position commands only on that explicit GO. Staging the command (first key) before the GO is fine; uplinking before it is a violation. Source: textbook Part 2, §9.2, *GO / NO-GO protocol*.

---

2. Procedure walk-through

2.1 NOM-001 Pass Startup

Textbook Part 2, §9.3, walks NOM-001 step by step against the MCS: the pre-checklist calculations, the checklist, the post-conditions, the common deviations, and which cards the MCS can run by itself. Use it as the session text, with `procedures/nominal/startup.md` open on the PRC page so students see each step become a line in the panel.

2.2 CTG-003 TTC Link Loss Recovery

Textbook Part 2, §9.4, is the walk-through: the escalation-ladder figure (check the ground station is tracking, blind-command the primary transponder, switch to the redundant one, retry the primary, verify, escalate to EMG-004), the read-and-do checklist, why it is structured that way, and what operators often miss. The loop to emphasise is detect (`ttc.link_status` , 0x0501, leaves 2) → assess (ground or space?) → recover (least intrusive rung first).

---

3. Shift handover practice (NOM-012)

Textbook Part 2, §9.5, gives the NOM-012 template and the drill. Each team performs two handovers: the outgoing FD briefs the incoming FD on spacecraft posture, open anomalies, work in progress and the next pass plan; the incoming FD reads it back verbatim; then authority is handed over on the loop. The instructor times each one. Target: under 6 minutes.

---

4. Flight-dynamics workshop

> **TODO (author):** Write the 400-word flight-dynamics workshop and its three Python figures (ground track with the five stations' visibility cones; elevation against time for a typical Iqaluit pass; eclipse fraction over a 24 h window). Exercises: read the TLE in `configs/eosat1/orbit.yaml` and compute the period (94.5 min) and revolutions per day (15.24); predict the next 24 h of Iqaluit AOS times; pass duration against maximum elevation (9.3 min overhead at 513 km with the 5° mask, 7.5 min on average); where eclipse starts, how long it lasts (33.2–35.2 min over the year), and its power and thermal consequences. Textbook Part 2, §9.6 and §§13.2–13.3, already have the arithmetic.

---

5. Dry-runs

Dry-run 1 — nominal pass

- **Scenario:** BP-1 (Nominal Post-Commissioning).
- **Duration:** 90 min real time, 90 min sim time (1× speed).
- **Procedures expected:** NOM-001 → NOM-002 (imaging) → NOM-003 (data downlink) → NOM-009 (routine health check) → LOS.
- **Failure injection:** none.
- **Outputs:** operator shift log; observer team uses *Communications* peer-review worksheet only.

Dry-run 2 — nominal pass with one failure

- **Scenario:** BP-1 with one easy failure injected ~T+30 min.
- **Recommended injection:** `eps_overcurrent.yaml` on the `payload` line (forces CTG-012 *Overcurrent Response*).
- **Operator pass criterion:** anomaly detected, isolated, and procedure CTG-012 executed; pass continues to LOS.
- **Outputs:** operator shift log + anomaly ticket; observer team uses *Technical* peer-review worksheet.

---

6. Operations training quiz

> **TODO (author):** Write the 15-question paper (multiple choice and short answer, 30 minutes), with solutions on a separate page for the instructor pack: voice-loop discipline (3 questions), MCS navigation (3), telemetry interpretation — limits and SIDs (3), procedure structure (2), anomaly response (2), and pass timing — AOS / TCA / LOS arithmetic (2).

---

7. Expected reading for Wednesday

> **Expected reading before Wednesday LEOP day.** Read sections 1 – 4 of the *LEOP Day Instructor Pack* student-facing extract (a redacted version of `sim_leop_instructor.md`). Approximate reading time 30 minutes. Read procedures **LEOP-001, LEOP-002, LEOP-006, LEOP-007** in full (`procedures/leop/*.md`). Approximate reading time 30 minutes.

---

8. Tools, files & references

Simulator files referenced

- `configs/eosat1/procedures/nominal/startup.md`
- `configs/eosat1/procedures/contingency/ttc_link_loss.md`
- `configs/eosat1/procedures/nominal/shift_handover.md`
- `configs/eosat1/procedures/leop/*.md` (for evening reading)
- `configs/eosat1/scenarios/eps_overcurrent.yaml`

Open references

- NASA APPEL — Voice loop and mission ops standards — https://appel.nasa.gov/
- ESA Operations *Mission Operations Concept* — https://www.esa.int/Enabling_Support/Operations
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- *NASA Systems Engineering Handbook* (SP-2016-6105 Rev 2) §8 — https://www.nasa.gov/reference/systems-engineering-handbook/
- CCSDS space packet standards (background; PUS itself is ECSS-E-ST-70-41C) — https://public.ccsds.org/Publications/AllPubs.aspx